

NetSentinel AI | Technoverse Hackathon 2026

Team: Udhbhav · Domain: Communications, Media & Technology · Theme: Network Automation

■ NetSentinel AI | Business Plan (1/2)

WHY

Explain the Problem

Problem Description & Business Scenario:

Telecom and communication networks in India and globally face rising incidents of physical infrastructure tampering, unauthorized access, and silent node failures. Traditional monitoring systems rely on periodic manual inspections and basic threshold alarms — both reactive and insufficient for modern network scale. According to GSMA Intelligence, global telecom infrastructure losses due to undetected faults and theft exceed \$25 billion annually. With India's 5G rollout adding 600,000+ new base stations by 2027, the need for intelligent, always-on, automated monitoring has never been more urgent.

Problem Scope:

Scope covers last-mile network nodes, telecom towers, distribution hubs, and IoT-enabled facilities. The problem is limited to physical infrastructure anomalies (tampering, vibration, weight shifts, magnetic field changes) and does not address cybersecurity threats. Solution targets single-site to multi-site deployments with centralized cloud monitoring.

Target Users / Stakeholders:

Primary: Telecom NOC (Network Operations Centre) teams, field maintenance engineers. Secondary: Utility companies, government infrastructure authorities, ISPs managing last-mile nodes. Pain points addressed: high MTTR (Mean Time to Repair), manual audit dependency, lack of real-time anomaly context, and inability to predict failures before they escalate.

Solution Overview:

NetSentinel AI is an IoT + Generative AI powered infrastructure monitoring system. ESP32 microcontrollers deployed at network nodes continuously read multi-sensor data (weight via HX711, vibration/tilt via MPU6500, magnetic field via HW-484). Data streams to Firebase Realtime Database. The cloud layer runs Google Gemini API to classify anomalies, generate natural language incident reports, and power a conversational operator assistant — all accessible via a real-time web dashboard without any app installation.

Technical Details:

Hardware: ESP32, HX711 load cell amplifier, MPU6500 6-axis IMU, HW-484 Hall effect sensor. Connectivity: WiFi → Firebase Realtime DB (JSON push, avg latency <200ms). Backend: Google Firebase (auth, DB, hosting). GenAI: Google Gemini 1.5 Flash API — multi-sensor context prompting for anomaly classification + NL report generation. Frontend: Single-file HTML5/JS/CSS with Chart.js for live graphs, AbortController-safe fetch, XSS-protected rendering. Languages: C++ (ESP32 firmware), JavaScript (dashboard), Python (data simulation & testing).

Innovation:

Unlike rule-based SCADA systems, NetSentinel AI moves from detection to understanding. The GenAI layer correlates multi-parameter sensor streams and produces human-readable diagnoses — explaining WHAT happened, WHY it likely happened, and WHAT action to take — autonomously, without expert intervention. This is a fundamentally different approach: AI as the first responder, not just an alert system.

Market Potential:

Global network infrastructure monitoring market: USD 4.8B (2024) → USD 11.2B by 2030 (CAGR 15.2%, MarketsandMarkets). India-specific opportunity: BSNL + private telcos 5G infrastructure capex of ₹1.5 lakh crore by 2027 creates direct demand for intelligent monitoring. IoT security market in India projected at USD 1.1B by 2026 (NASSCOM).

Why these technologies are the right fit:

ESP32 is the industry standard for low-cost, WiFi-enabled edge sensing — widely deployed in telecom and industrial IoT. Firebase enables zero-infrastructure-overhead real-time data sync, ideal for rapid deployment across distributed nodes. Gemini API provides state-of-the-art multimodal reasoning at low cost, making AI-powered diagnostics economically viable even for small telecom operators.

How

Explain the Solve

WHAT

Value Proposition

Primary Benefits:	60% reduction in Mean Time to Repair (MTTR); zero missed anomalies via 24/7 AI watch
Efficiency & Flexibility:	Auto-generated incident reports eliminate manual log analysis; dashboard accessible from any browser, any device
Time & Cost Saving:	Estimated ₹8–12 lakh/year savings per 100 nodes by eliminating unnecessary field dispatches; prototype hardware cost <₹2,500 per node
Scalability:	Firebase architecture scales to 100,000+ concurrent nodes; modular sensor design allows addition of new sensor types without firmware overhaul
Social Impact:	Ensures reliable connectivity for underserved rural communities; prevents telecom revenue leakage that funds network expansion into Tier 2/3 cities

Financials & Timelines | Business Plan (2/2)

Investments

What does it take &
How much does it
cost?

Cloud (Firebase)	Blaze plan — ₹0 up to 1GB storage, ~₹2,000/month for 10,000 nodes
AI (Gemini API)	Gemini 1.5 Flash: \$0.075/1M tokens; est. ₹1,500/month for 500 nodes active
Edge Hardware	₹2,200–2,500 per node (ESP32 + sensors + enclosure); one-time cost
Process Re-engg	NOC workflow redesign: 2 weeks; operator training: 4 hours per technician
Modernization	Replaces legacy SCADA dashboards; REST API bridges to existing NMS systems
Tools	VS Code, Arduino IDE, Firebase Console, Postman, Gemini API Studio
Total Pilot Cost (50 nodes)	~₹1.5 lakh hardware + ₹25,000/year cloud + AI — breakeven in <3 months via savings

ROI

300%+ ROI within 12 months for a 100-node deployment; hardware cost recovered in <3 months via reduced field dispatch

Revenue Growth

Telecom operators recover ₹10–15 lakh/year per district in prevented SLA penalty payments and theft-related losses

Happy Customer

End-user network uptime improves by 35–40%; faster fault resolution = fewer customer complaints and churn

Margin Improvement

60% reduction in truck rolls (unnecessary field visits); NOC staff productivity up 3x via AI-generated diagnostics

Employee Experience

Field technicians receive precise AI-guided repair instructions; NOC analysts freed from manual log reviews

If NOT solved

Every undetected node failure costs avg ₹45,000 in SLA penalties + ₹80,000 in emergency repair; with 5G densification, failures will only multiply without AI-driven monitoring

Returns

Quantify benefits &
What if I don't solve?

Timelines

Time to realize
benefits

Milestone	Timeline	Deliverable
Phase 1 — Prototype	0–4 weeks	Working ESP32 + Firebase + Gemini integration on 5 nodes
Phase 2 — Pilot	1–3 months	50-node pilot with NOC dashboard; SLA improvement measured
Phase 3 — Scale	3–6 months	500+ node rollout; NMS/OSS API integration; ROI realized
Phase 4 — Enterprise	6–12 months	Multi-tenant SaaS; TinyML edge inference; compliance reporting
Time to first benefit	Week 3	First AI-generated anomaly report replacing manual audit